

This feedback is machine-assisted; every point must be verified against the paper before acting on it.

Feedback on: Physics-Informed but Not Physics-Consistent: Error Geometry and Subspace Projection for Neural AC Power Flow

Summary

The paper investigates why neural AC power-flow surrogates can have small voltage-state errors while retaining large power-balance residuals. It decomposes prediction errors relative to a training-derived SVD solution subspace, uses a fixed-norm intervention to test the role of error direction, and proposes calibrated solution-subspace projection (CSP) as a train-only output correction. Experiments cover four neural models on a 2224-bus GBnetwork, GridSFM on 31 systems, and selected topology and spatial distribution shifts.

Recommendation

Major revision — the empirical result is promising and the central diagnostic is suitable for a focused ICASSP paper, but the current manuscript leaves the actual projection operator underdefined, does not report the evidence needed for its two central mechanistic claims, and positions the method too broadly against existing work on power-flow geometry, PCA output spaces, and feasibility correction. Before submission, the paper should define the magnitude and angle projectors unambiguously, quantify off-subspace error, narrow the causal and generality claims, document the baseline and data protocols, and repair the cross-grid rank inconsistency. Most problems are fixable within the existing study, but a fair residual-directed correction baseline and stronger cross-grid reporting would materially improve the paper.

1. Literature & Related Work

Positioning and novelty assessment

The strongest contribution is not the existence of a low-dimensional solution space or the general idea of post-processing predictions. It is the combination of:

- a train-derived coordinate system for decomposing errors of already trained neural AC-PF surrogates;
- a fixed-norm intervention that varies in-subspace and off-subspace error orientation;
- a lightweight post-training composition of per-bus bias calibration and linear solution-subspace projection;
- evaluation across four heterogeneous model families on one large system.

The current positioning overstates this contribution in three places:

- Abstract: “output-error geometry as a critical factor.”
- §1: “the missing factor lies in the geometry.”
- §4: “identifying output-error geometry as a key source.”

The experiments establish that error direction affects residuals in the evaluated models. They do not establish that geometry is the unique, principal, or “critical” explanation relative to total error magnitude, Jacobian conditioning, bus type, topology, operating-point difficulty, or proximity to voltage collapse. Replace these formulations with “an underexamined factor” or “an important factor in the evaluated surrogates.”

CSP should be positioned as an approximate, model-agnostic, linear output correction. The paper correctly states in §2.3 that CSP is not an AC-feasible-set projection and does not guarantee exact feasibility. That distinction should also be made in the related-work paragraph, where exact feasibility restoration, iterative residual correction, equality completion, and learned approximate projection currently appear as one undifferentiated category.

Citation audit: problems found

- Citation [22], Park et al., is closer prior art than §1 conveys. The checked source, “Compact Optimization Learning for AC Optimal Power Flow” (arXiv:2301.08840), uses PCA to compress AC-OPF solution

outputs, learns within the principal-component subspace, reconstructs full outputs, and uses them to warm-start an exact solver for feasibility restoration. The current sentence mentions only compression. Expand it and distinguish CSP by its post-training application to arbitrary frozen models, fixed-norm error intervention, separate bias correction, and diagnostic purpose.

- Citation [15] documents PIGNN-Attn-LS, not PIGNN-GC. The checked source (arXiv:2509.22458) defines edge-aware attention and a backtracking line-search correction operator. §3.1 calls the evaluated method “a global-context extension” but gives no architecture or reference for that extension. Define PIGNN-GC and state whether the published line-search operator remains active.
- Citation [18] is not reproducible as written. The entry says “Software documentation, 2026, Version 0.9.0; official citation pending” but provides no URL, commit, checkpoint, or access date. Add an immutable software identifier.
- GridSFM [17] and LUMINA [20] are introduced as AC-OPF foundation models, but §3.1 does not explain how their inputs and outputs are converted into the AC-PF state-prediction experiment. State which outputs are retained, whether either model was fine-tuned, how generator variables are treated, and which checkpoints were used.
- PowerFlowNet [11] is cited as a direct neural PF surrogate but omitted experimentally. Its published identifier is DOI: 10.1016/j.ijepes.2024.110112. Either include it or state that the experiment is a post-processing study of selected frozen outputs rather than a comprehensive learned-solver benchmark.

Missing references

- Goodwin, Maack, and Sigler, “Power Flow Geometry and Approximation,” *IEEE Transactions on Power Systems*, 2026, DOI: 10.1109/TPWRS.2025.3612220. It directly studies nonlinear AC-PF solution-manifold geometry. Cite it in §1 immediately before “the missing factor lies in the geometry” and in §2.2; distinguish formal PF-manifold geometry from the empirical geometry of neural errors relative to a linear training subspace.
- Rivera et al., “PF Δ : A Benchmark Dataset for Power Flow under Load, Generation, and Topology Variations,” NeurIPS 2025, arXiv:2510.22048. It contains 859,800 solved instances across six system sizes, including N , $N - 1$, $N - 2$, and close-to-infeasible cases. Cite it in §1 and §3.1; ideally evaluate CSP on it or explain why the custom GBnetwork study addresses a different question.
- Donti et al., “DC3: A Learning Method for Optimization with Hard Constraints,” ICLR 2021, arXiv:2104.12225. It uses completion and correction to satisfy equality and inequality constraints, including AC-OPF. Cite in §1’s correction-method taxonomy.
- Chzhen and Donti, “Improving Feasibility via Fast Autoencoder-Based Projections,” 2026, arXiv:2604.03489. It is a plug-and-play learned approximate projector for nonconvex feasible sets. Cite in §1 and contrast its learned nonlinear feasible-set approximation with CSP’s linear training-solution projection.
- “DeepOPF: A Feasibility-Optimized Deep Neural Network Approach for AC Optimal Power Flow Problems,” arXiv:2007.01002. It predicts independent variables and completes the solution through power-flow computation. Cite in §1 alongside correction and completion methods.
- Nellikkath and Chatzivasileiadis, “Physics-Informed Neural Networks for AC Optimal Power Flow,” *Electric Power Systems Research* 212, 108412, DOI: 10.1016/j.epsr.2022.108412. Cite in §1’s physics-informed objective discussion.
- Chevalier and Chatzivasileiadis, “Global Performance Guarantees for Neural Network Models of AC Power Flow,” arXiv:2211.07125. Cite after the discussion of average accuracy versus consistency and revisit in §3.3 when discussing extreme topology-shift residuals.
- OPFData (arXiv:2406.07234), PGLib-OPF (arXiv:1908.02788), and pandapower (arXiv:1709.06743) should be cited in §3.1 or §3.3 wherever the model-training data and 31 network cases originate. Exact case and software versions must accompany the citations.

2. Mathematical Correctness

Undefined or used-before-defined symbols

- §2.1 uses V for voltage magnitude and visually similar V for the complex voltage phasor: $> “x = [V, \theta]$, where $V = |V|$ and $V = V \odot e^{j\theta}$.”

Replace this with distinct symbols, for example

$$x = [v^\top, \theta^\top]^\top, \quad \mathbf{V} = v \odot e^{j\theta}.$$

- I denotes current in §2.1, $I = Y\mathbf{V}$, and the identity operator in Eq. (2), $e_\perp = (I - P_k)e$. Use $\mathbf{i}$ for current and I_{2N} , I_N , or another dimensioned symbol for identity.
- Eq. (1) introduces N_{tr} through the number of columns but never explicitly defines it as the number of training scenarios.
- Eq. (1) calls P_k block diagonal, but the magnitude and angle data matrices, their individual ranks, and their individual singular-vector matrices are never defined.
- Eq. (5) uses the subscript w in R_w^2 and a_w without explaining what w denotes.
- Eq. (5) uses i, s but does not give their ranges or state whether covariance and correlation are computed globally over all bus-scenario pairs, separately by bus and then averaged, or using another weighting.
- Eq. (6) should explicitly state $\alpha \in [0, 1]$.
- P^{set} and Q^{set} are introduced in §2.1 without defining their availability at slack, PV, and PQ buses. The later phrase “standard slack/PV/PQ conventions” does not define the actual masks.
- The terms AC, SVD, PCA, p.u., PV, PQ and RMSE are never expanded. These are addressed further in §3 below.

Errors and gaps in derivations

The orthogonal decomposition in Eq. (2) is correct if $P_k = P_k^\top = P_k^2$. Eq. (1) should state those properties after defining the actual block projector.

The first-order Taylor expansion in Eq. (3) is locally correct under a fixed bus-type regime and differentiability of the residual map. The paper must state that it is a local argument and that reactive-limit switching, topology changes, ill-conditioned Jacobians, and voltage-collapse regimes can invalidate or weaken the approximation. No theorem or global consistency result follows from Eq. (3).

Eq. (3) shows that residual amplification depends on Je ; it does not show that off-subspace errors have higher Jacobian gain. Add at least one of:

$$\|JP_k e\|_2, \quad \|J(I - P_k)e\|_2,$$

restricted singular-value summaries, or principal-angle analysis. Without this, the mathematical mechanism invoked by the paper is not tested directly.

Equation (1) must be rewritten to match the implementation. If magnitude and angle use separate bases, define

$$X_v = [v_1^* - \bar{v}, \dots, v_{N_{\text{tr}}}^* - \bar{v}], \quad X_\theta = [\tilde{\theta}_1^* - \bar{\theta}, \dots, \tilde{\theta}_{N_{\text{tr}}}^* - \bar{\theta}],$$

$$X_v = W_v \Sigma_v U_v^\top, \quad X_\theta = W_\theta \Sigma_\theta U_\theta^\top,$$

$$P_{k_v, k_\theta} = \text{diag} \left(W_{v, k_v} W_{v, k_v}^\top, W_{\theta, k_\theta} W_{\theta, k_\theta}^\top \right).$$

State whether $k_v = k_\theta = 16$, whether 16 is a total rank, and how angles are aligned.

The cross-grid rank must be corrected. Figure 3 applies CSP_{16} to systems as small as four buses. Define

$$k_{v,g} = \min\{16, \text{rank}(X_{v,g}), N_g\}, \quad k_{\theta,g} = \min\{16, \text{rank}(X_{\theta,g}), N_g - 1\},$$

or report the different rule actually used.

Equation (6) is undefined when its normalization denominator is zero. This includes exact predictions and pure in-subspace or off-subspace errors at $\alpha = 0$. State an exclusion or limiting-value convention and report how often it is invoked.

Equation (8) is undefined if $\text{Mean PB}(1) = 0$. State that exact-consistency cases are reported separately rather than ratio-normalized.

Equation (4) is an unconstrained affine projection and can produce nonpositive or out-of-range voltage magnitudes and alter fixed slack/PV setpoints. Add voltage-bound and setpoint checks, or specify a post-projection clipping/rejection convention. Report whether CSP changes generator, thermal, or voltage-limit violations.

The PB definition is incomplete. Give explicit masks m_P, m_Q and define, for example,

$$\text{PB}_{is} = \sqrt{(m_{P,i} \Delta P_{is})^2 + (m_{Q,i} \Delta Q_{is})^2},$$

with m_P, m_Q specified for slack, PV, and PQ buses. Also define whether Mean PB is an equally weighted mean over buses and scenarios.

The k^* selection rule in §3.1 is not operational:

“validation Mean PB selects $k^* = 16$ without degrading calibrated voltage RMSE.”

Define “without degrading,” the tie-break rule, whether selection is performed per model and seed, and the aggregation used across models. Include the validation curve for $k \in \{4, 8, 16, 32, 64\}$.

Overall soundness

The algebra of calibration in §2.1 is sound for a frozen empirical training-set mean. The stated invariance of centered R_w^2 and a_w under additive per-bus calibration is also correct. The orthogonal decomposition and fixed-norm construction are sound once a valid orthogonal projector is specified and zero-denominator cases are excluded.

The paper proves no theorem, exact feasibility result, bound, convergence property, or global consistency guarantee. Its contribution is empirical. The mathematical claims should remain local and diagnostic.

The equations were checked algebraically from the manuscript. The implementation, repository code, raw scenarios, trained checkpoints, SVD outputs, and table-generation scripts were not part of this mathematical check.

3. Spelling, Grammar, Acronyms

Acronyms used before or never defined

Eight standard technical abbreviations are never expanded:

- AC — first used in the title.
- SVD — first used in the abstract.

- PCA — first used in §1.
- p.u. — first used in the abstract.
- PV — first used in §2.1.
- PQ — first used in §2.1.
- RMSE — first used in §3.1/Table 1.
- RTE — first used in §3.3.

Four method-name components are also unexplained:

- PIGNN.
- GC in PIGNN-GC.
- Attn in PIGNN-Attn-LS.
- LS in PIGNN-Attn-LS.

AC-PF, CSP and CIM are defined. Model names such as GridSFM and LUMINA can remain formal proper names, but PIGNN-GC must be described because it is an unpublished modification.

Typos and errors

- §3.2: Table 3 compares → Table 1 compares.
- Table 2 caption: `seperate` → `separate`.
- Table 2 caption: `quntiles` → `quantiles`.
- §1: `respectively. [10].` → `respectively [10].`
- Figure 1: `SVD k rank basis` → `rank- $\backslash(k\backslash)$  SVD basis`.
- Figure 1: `gridfm` → `GridFM-GraphKit`.
- Figure 1: `Physics Informed` → `Physics-informed`.
- Figure 1: `Reference train AC-PF` → `Training-set AC-PF reference solutions`.
- Figure 1: `Reduced Small voltage Error and Residual` → `Reduced voltage error and power-balance residual`.
- Table 2: `RMSEV` → `$\backslash(\mathrm{RMSE}_{|V|})$` to match Table 1.

Consistency and cross-reference issues

- The §3.2 reference to Table 3 does not resolve; there is no Table 3.
- Figure 1 uses `gridfm`, while the text and tables use `GridFM-GraphKit`.
- Table 1 uses `|V| RMSE`; Table 2 uses `RMSEV`.
- Use `power-balance residual` adjectivally and `power balance` nominally; the manuscript currently alternates among “power balance,” “power-balance,” “PB,” and “physics violation.”
- Define whether `GBnetwork` is the formal case name. If not, expand GB at first use.
- State the system MVA base used for PB. Cross-grid p.u. comparisons in Figure 3 are not interpretable without a consistent base convention.
- Figure 2’s caption calls the increase “complementary.” The two norm-renormalized interventions are complementary controls, not algebraic complements. Replace with “the norm-matched control suppressing in-subspace error increases PB.”
- Figure 1 should label the two correction stages as C and P_k to match Eq. (4) and Table 1.
- Use distinct symbols v and $\mathbf{V}$ throughout to separate magnitude from complex voltage.
- The manuscript consistently uses US English in the main prose; retain `analyzing`, `characterize`, `normalized`, and `toward`.

4. Reproducibility

Area	Status	Evidence and required repair
Data	Partial	§3.1 names a 2224-bus GBnetwork and 37,022 converged scenarios with an approximate 1:1:1 split. Missing are case source/version, exact split counts and indices, scenario distributions and ranges, solver implementation and tolerances, rejection rate, normalization, and all 31 cross-grid case names. Shift-set sizes and generators are also missing.
Method	Partial	Eqs. (1)–(4) describe calibration and CSP conceptually. Missing are separate magnitude/angle projectors, rank allocation, scaling, gauge alignment, effective-rank handling, SVD tolerance, bounds handling, and complete PIGNN-GC architecture.
Protocol	Partial	PB, R_w^2 , a_w , RMSE and validation rank selection are named, and test-set isolation is stated. Missing are explicit bus masks, aggregation denominators, exact rank-selection rule, seed semantics, number of runs outside Table 1, comparable baseline tuning, and checkpoint selection.
Compute	Missing	Hardware, software/library versions, model versions, checkpoints, training time, inference time, SVD fitting time, CSP overhead, memory, and parameter counts are absent.
Statistics	Partial	Table 1 reports mean $\pm$ SD over three seeds. Figure 2, Figure 3 and Table 2 report no variance. No paired effect size, confidence interval, scenario bootstrap, or grid-level uncertainty is provided.
Code release	Partial	§1 gives a GitHub URL. The paper gives no commit, tag, archive DOI, license, artifact manifest, or mapping from scripts to figures and tables.
Data release	Missing	No release statement covers the 37,022 scenarios, split indices, distribution shifts, or cross-grid datasets.

Area	Status	Evidence and required repair
Pretrained models	Missing	No checkpoint identifiers or adaptation instructions are given for GridSFM, GridFM-GraphKit, LUMINA, or PIGNN-GC.
Baseline fairness	Missing	No common data, compute, hyperparameter-search, validation, early-stopping, or inference-budget protocol is stated.

Verdict: A competent peer cannot reproduce the central numerical result from the paper alone.

Blocking items, in order:

1. Exact definition of the magnitude and angle projectors, including gauge, scaling, rank, and small-grid handling.
2. Exact GBnetwork scenario generator, reference-solver settings, split indices, and preprocessing.
3. Model checkpoints and full PIGNN-GC specification.
4. Bus-type residual masks and metric aggregation.
5. Exact identities, versions, and scenario counts for all 31 systems.
6. Training, adaptation, tuning, and checkpoint-selection procedure for every model.
7. Distribution-shift generation and sample sizes.
8. Seeds, software versions, and hardware.
9. A tagged or archived code release and a clear data-release statement.

5. Strengths

1. **The paper separates state accuracy from physical consistency explicitly.** Section 2.1 recomputes the AC residual from each final predicted voltage state instead of relying on training losses, and Table 1 shows that small magnitude RMSE can coexist with large Mean and Max PB.
2. **The fixed-norm intervention is a strong diagnostic design.** Equations (6)–(8) attempt to isolate error direction from magnitude, and Figure 2 includes the appropriate opposite-direction control rather than reporting only off-subspace suppression.
3. **The paper distinguishes the oracle diagnostic from the deployable correction.** Section 3.2 explicitly states that the controlled intervention requires x^* , while Eq. (4) uses only frozen training statistics at test time.
4. **Test-set leakage is addressed clearly.** Sections 2.1, 2.3, and 3.1 assign calibration and basis fitting to training data and rank selection to validation data before freezing the test protocol.
5. **The paper recognizes that low PB alone can be misleading.** Equation (5) introduces scenario-tracking measures, and Table 1 includes a per-bus mean baseline with $R_w^2 = a_w = 0$. This is an important control for foundation-model outputs.
6. **The principal GBnetwork reductions are large.** Table 1 reports Mean PB reductions from 0.31592 to 0.05497 for PIGNN-GC and from 0.26867 to 0.05136 for GridSFM. These are substantially larger than the displayed three-seed standard deviations.
7. **The distribution-shift section exposes rather than hides failures.** Table 2 shows extreme Max PB under $N - 1$ and $N - 2$ and only partial tracking recovery under GBcorr.
8. **The paper does not falsely claim exact feasibility.** Section 2.3 explicitly distinguishes CSP from projection onto the nonlinear AC-feasible set and disclaims exact AC-equation and operational-constraint guarantees.

6. Weaknesses

Major

1. **The method is underdefined at its mathematical core.** Equation (1) shows one SVD but describes separate block projectors; magnitude/angle scaling, gauge alignment, rank allocation, and the effective rank below 16 are missing. This is fixable in revision by replacing Eq. (1) with explicit X_v , X_θ , and block-projector definitions.
2. **The two central mechanistic claims are not backed by the reported measurements.** The Abstract and §1 call off-subspace error “substantial” and “strongly associated” with PB, but no off-subspace energy or association statistic is reported. Add $\|e_\perp\|^2/\|e\|^2$, an explained-variance spectrum, and a conditional association analysis.
3. **The Jacobian explanation is incomplete.** Equation (3) establishes local directional dependence but does not show that off-subspace directions have greater residual gain. Add $JP_k e$ versus $J(I - P_k)e$ analysis or narrow the mechanistic claim.
4. **The nearest prior art is insufficiently distinguished.** Section 1 underdescribes Compact Optimization Learning (arXiv:2301.08840) and omits direct PF geometry and projection/correction literature. This is fixable through tighter positioning and citations; the method should not be sold as the first solution-subspace or feasibility-projection approach.
5. **Baseline comparisons are not fair or reproducible.** Section 3.1 gives no common training, tuning, checkpoint, or compute protocol, and PIGNN-GC is not defined by citation [15]. Add a compact configuration table and characterize the study as post-processing fixed model outputs.
6. **There is no matched residual-directed correction baseline.** Table 1 needs at least one Newton/residual-gradient step or the PIGNN line-search correction at reported matched cost. Without it, CSP’s practical competitiveness is unknown.
7. **Cross-grid conclusions exceed cross-grid evidence.** Figure 3 reports only GridSFM Mean PB, while §4 claims reductions in voltage error and PB across diverse grids. Either add cross-grid voltage/tracking/tail metrics or narrow the conclusion.
8. **The cross-grid rank is internally inconsistent.** CSP_{16} cannot literally be rank 16 on four-bus grids. Add the effective-rank convention.
9. **Distribution-shift adaptation is confounded.** Section 3.3 simultaneously refits bias calibration and the basis for GBcorr, then attributes improvement to basis adaptation. Add the four calibration/basis transfer-refit combinations or change the interpretation.
10. **Operational consistency is not measured.** Section 2.3 disclaims operational guarantees, but the title and conclusions motivate physics consistency broadly. Report feasibility rates, voltage/setpoint violations, and preferably branch/generator limits, or define the paper’s scope strictly as nodal PB consistency.
11. **The data protocol blocks independent verification.** Section 3.1 lacks the scenario distributions, split indices, solver tolerances and preprocessing; §3.3 does not identify the 31 systems. These are fixable through a compact appendix or archived repository manifest.

Minor

1. Table 1 shows that CSP slightly worsens angle RMSE relative to calibration for PIGNN-GC and GridSFM. Change “improves voltage accuracy” to “improves voltage-magnitude accuracy” when comparing CSP with calibration.
2. GridFM-GraphKit’s calibrated Max PB increases from 4651.17 to 4853.45 in Table 1. Mention this explicitly because it reinforces the distinction between mean accuracy and worst-case consistency.

3. The four GridSFM $a_w = .724$ entries in Table 1 should be audited. Calibration explains equality within Raw/C and P/CSP pairs, but projection does not generally preserve a_w .
4. Table 2 pools all bus–scenario pairs, overweighting larger systems and failing to describe scenario-level operational risk. Add the distribution of per-scenario Max PB.
5. Equation (6) and Eq. (8) need zero-denominator conventions; Eq. (1) needs numerical-rank and tied-singular-value conventions.
6. Section 3.1’s rank-selection statement needs a numerical non-degradation tolerance and tie-break rule.
7. Figure 2, Figure 3 and Table 2 need run counts and uncertainty. A paired scenario bootstrap nested within seed is sufficient.
8. The paper evaluates only converged Newton–Raphson cases. Report the discarded-case rate and add conditioning or distance-to-collapse stratification if available.
9. The Abstract’s “particularly on large-scale grids” is unsupported by a controlled scale analysis. Remove “particularly” or add a size-effect analysis.
10. The §1 claim that useful geometry is “low-dimensional” needs a singular-value spectrum, variance explained, and rank-sensitivity curve.
11. Fix the nonexistent Table 3 reference, the two Table 2 typos, the Figure 1 labels, undefined abbreviations, metric naming, and voltage-symbol collision.

7. What to fix before 17 September

1. Rewrite Eq. (1) and Eq. (4) with separate magnitude and angle matrices, explicit gauge alignment, scaling, rank allocation, SVD tolerance, and k_{eff} for small grids.
2. Add one compact quantitative table or panel reporting off-subspace error fraction, explained variance, and the relationship between $\|e_{\perp}\|$ and PB conditional on total error.
3. Add a Jacobian-gain check or remove the stronger mechanistic language derived from Eq. (3).
4. Narrow “critical factor,” “the missing factor,” “key source,” “low-dimensional,” “particularly on large-scale grids,” and the cross-grid voltage claim to what the experiments demonstrate.
5. Expand related work with PF geometry (DOI: 10.1109/TPWRS.2025.3612220), PF Δ (arXiv:2510.22048), Compact Optimization Learning (arXiv:2301.08840), DC3 (arXiv:2104.12225), and FAB (arXiv:2604.03489).
6. Add one matched-cost residual-directed correction baseline, preferably the existing PIGNN line-search correction or a fixed number of Newton/residual-gradient steps.
7. Add a reproducibility table giving data generation, exact splits, model checkpoints, PIGNN-GC definition, training/adaptation settings, residual masks, 31 case names, software versions, seeds, and runtime.
8. Split GBcorr recalibration from basis refitting, or state only that joint condition-matched refitting improves PB.
9. Either add cross-grid voltage/tracking/tail results or restrict the conclusion to cross-grid Mean PB reduction for GridSFM.
10. Report scenario-level tail risk and operational checks, especially for the $N - 1/N - 2$ cases with Max PB of 33.16 and 27.45.
11. Freeze the repository with a tag or archive, state what data will be released, and map each figure/table to an executable script.
12. Complete the editorial pass: Table 3 $\rightarrow$ Table 1; **seperate** $\rightarrow$ **separate**; **quntiles** $\rightarrow$ **quantiles**; redraw Figure 1; define the eight standard abbreviations; standardize method names, units, and metric notation.